General Characteristics

21. The catalytic activity of the transition metals and their compounds is ascribed to their

(a) Chemical reactivity

(b) Magnetic behaviour

(c) Unfilled *d*-orbitals

(d) Ability to adopt multiple oxidation states and their complexing ability

22. What is the general electronic configuration for 2nd row transition series

(a) $[Ne]3d^{1-10}, 4s^2$

(b) $[Ar]3d^{1-10}, 4s^{1-2}$

(c) $[Kr]4d^{1-10}, 5s^{1-2}$

(d) $[Xe]5d^{1-10}, 5s^{1-2}$

23. Transitional elements are named transition elements because their characters are

(a) In between *s* and *p* - block elements

(b) Like that of *p* and *d* - block elements

(c) They are members of *I – A* group

(d) They are like inactive elements

24. Those elements whose two outermost orbitals are incompletely filled with electrons are

(a) *p* - block elements

(b) *s* - block elements

(c) Transitional elements

(d) Both *s* and *p* - block elements

25. Which ion has maximum magnetic moment

(a) V^{+3} (b) Mn^{+3}

(c) Fe^{+3} (d) Cu^{+2}

26. Which of the following transition metal is present in misch metal

(a) La (b) Sc

(c) Ni (d) Cr

27. Which of the following statements is not true in regard to transition elements

(a) They readily form complex compounds

(b) They show variable valency

(c) All their ions are colourless

(d) Their ions contain partially filled *d* -electron levels

28. Which of the following represents the electronic configuration of a transition element

(a) $1s^2, 2s^2p^6 .... ns^2p^3$

(b) $1s^2, 2s^2p^6 .... ns^2p^6d^3, (n+1)s^2$

(c) $1s^2, 2s^2p^6 .... ns^2p^6d^{10}, (n+1)s^2p^1$

(d) $1s^2, 2s^2p^6 .... ns^2p^6$

29. The general electronic configuration of transition elements is

(a) $(n-1)d^{1-5}$

(b) $(n-1)d^{1-10}ns^1$

(c) $(n-1)d^{1-10}ns^{1-2}$

(d) $ns^2(n-1)d^{10}$

30. Transition elements are coloured

(a) Due to small size
(b) Due to metallic nature
(c) Due to unpaired *d* - electrons
(d) All of these

31. Which of the following has the maximum number of unpaired *d*-electrons
(a) Zn (b) Fe^{2+}
(c) Ni^{3+} (d) Cu^{+}

32. Which is not amphoteric
(a) Al^{3+} (b) Cr^{3+}
(c) Fe^{3+} (d) Zn^{2+}

33. Which does not form amalgam
(a) Fe (b) Co
(c) Ag (d) Zn

34. Transition metals are often paramagnetic owing to
(a) Their high M.P. and B.P.
(b) The presence of vacant orbitals
(c) The presence of one or more unpaired electrons in the system
(d) Their being less electropositive than the elements of groups I-A and II-A

35. Elements which generally exhibit multiple oxidation states and whose ions are usually coloured
(a) Metalloids
(b) Transition elements
(c) Non-metals
(d) Gases

36. Which of the following transition metal cation has maximum unpaired electrons
(a) Mn^{+2} (b) Fe^{+2}
(c) Co^{2+} (d) Ni^{2+}

37. Maximum number of oxidation states of transition metal is derived from the following configuration
(a) *ns* electron
(b) $(n-1)d$ electron
(c) $(n+1)d$ electron
(d) $ns+(n-1)d$ electron

38. Which of the following statement is correct
(a) Iron belongs to 3rd transition series of the periodic table
(b) Iron belongs to *f*-block of the periodic table
(c) Iron belongs to second transition series of the periodic table
(d) Iron belongs to group VIII of the periodic table

39. Zinc does not show variable valency like *d*-block elements because]
(a) It is a soft metal
(b) *d*-orbital is complete
(c) It is low melting
(d) Two electrons are present in the outermost orbit

40. Which of the following is a transitional element
(a) Al (b) As
(c) Ni (d) Rb

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